

Lecture 3:

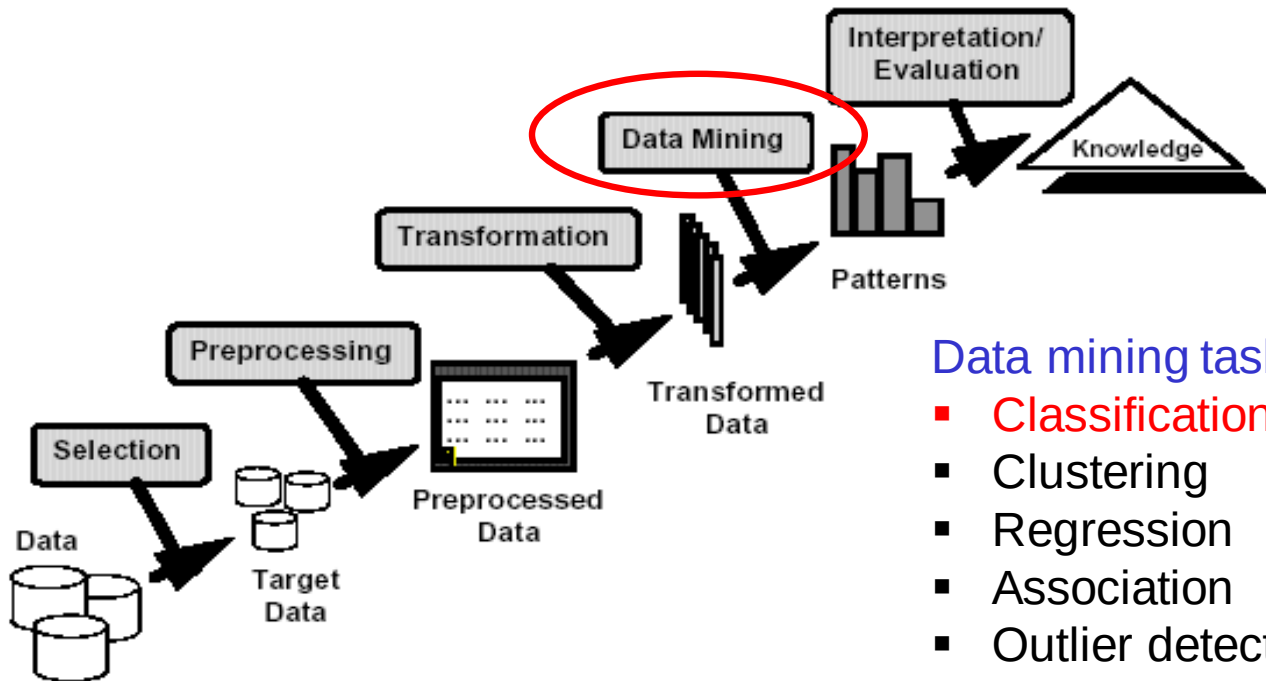
Data classification (I)

Outline

- Motivation
- Classification models – basic concepts
- Performance measures
- Classifiers
 - Simple voting (ZeroR)
 - Simple classification rules (OneR)

Motivation

Reminder: knowledge discovery workflow



Data mining tasks:

- Classification
- Clustering
- Regression
- Association
- Outlier detection

Motivation

Reminder: examples of classification problems

- Predicting tumor cells as benign or malignant (**medical diagnosis**)
- Classifying credit card transactions as legitimate or fraudulent (**fraud detection**)
- Categorizing news stories as finance, weather, entertainment, sports, etc (**document classification**)
- Classifying e-mails as spam or ham (**spam filtering**)
- other examples ...

Motivation

- **Medical diagnosis** = predict the presence/absence of an illness based on characteristics available in a health record

Example of a data subset (breast-cancer-wisconsin - arff format – see Lab 1)

```
@relation wisconsin-breast-cancer
@attribute Clump_Thickness integer [1,10]
@attribute Cell_Size_Uniformity integer [1,10]
@attribute Cell_Shape_Uniformity integer [1,10]
@attribute Marginal_Adhesion integer [1,10]
@attribute Single_Epi_Cell_Size integer [1,10]
@attribute Bare_Nuclei integer [1,10]
@attribute Bland_Chromatin integer [1,10]
@attribute Normal_Nucleoli integer [1,10]
@attribute Mitoses integer [1,10]
@attribute Class { benign, malignant}
@data
5,1,1,1,2,1,3,1,1,benign
5,4,4,5,7,10,3,2,1,benign
3,1,1,1,2,2,3,1,1,benign
8,10,10,8,7,10,9,7,1,malignant
1,1,1,1,2,10,3,1,1,benign
```

Predictive (input) attributes

Class (output) attribute

Two classes = binary classification

Basic concepts

What is known?

- a collection of records for which it is known to which class they belong (**labeled data set**)
- each record contains a set of **attributes**, one of the attributes is the **class label**

What is desired?

- a **model** which captures the relationship between the class attribute and the other attributes (the model is inferred using a **training set** through a process which is called **supervised learning/ training**)

Which is the final aim?

- to use the inferred model to identify the right class for a previously unseen record

Remarks:

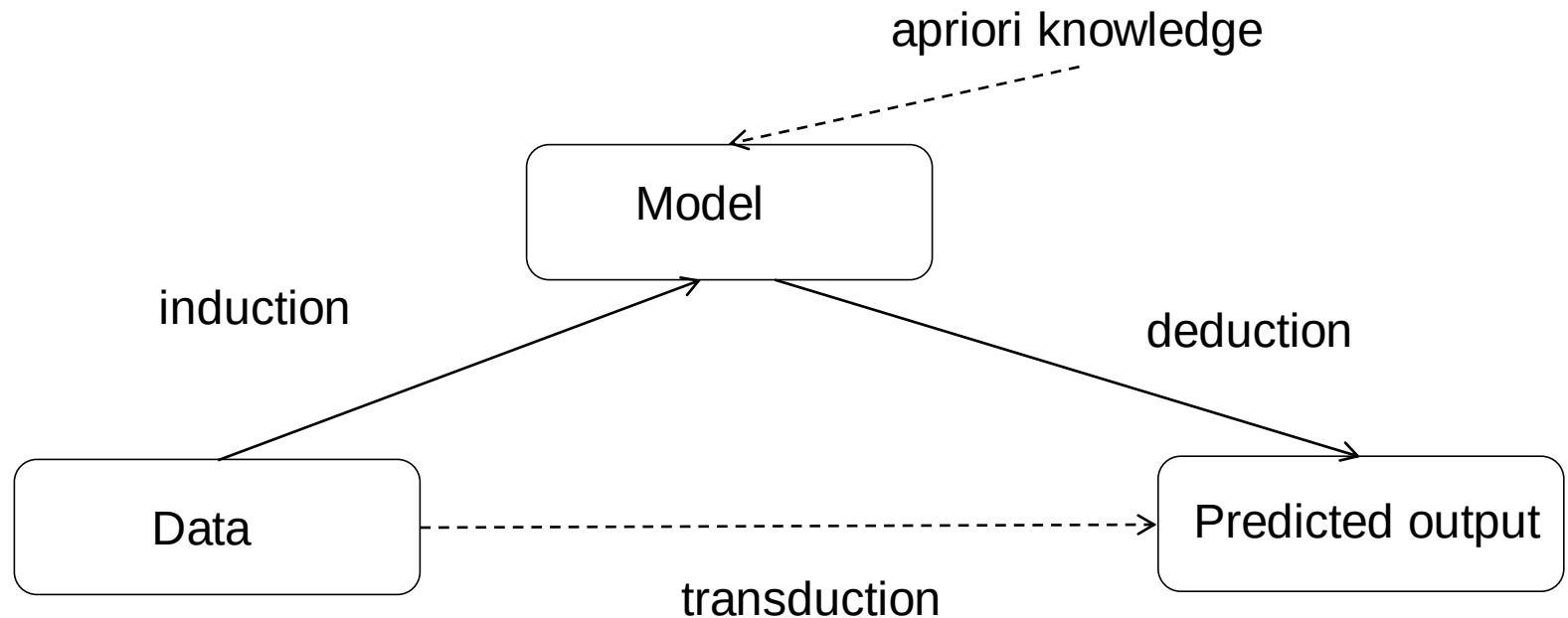
- a useful model should be **accurate**; the model accuracy/performance can be estimated using the data which have not been used during the learning (**testing set**)

Basic concepts

Learning/ induction/ inference = construct a model starting from data (and some apriori knowledge specific to the domain)

Different ways of using data, models and knowledge:

induction vs deduction vs transduction

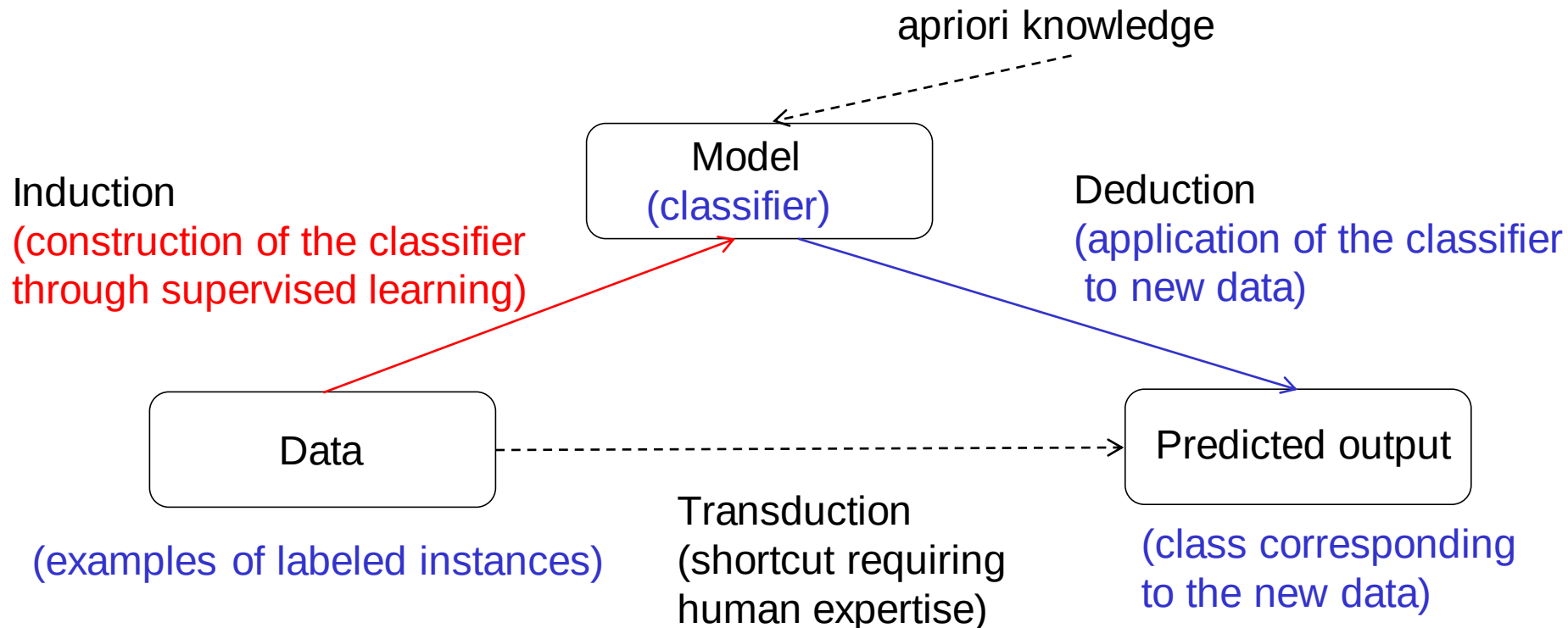


Basic concepts

Learning/ induction/ inference = construct a model starting from data (and some apriori knowledge specific to the domain)

Different ways of using data, models and knowledge:

induction vs deduction vs transduction



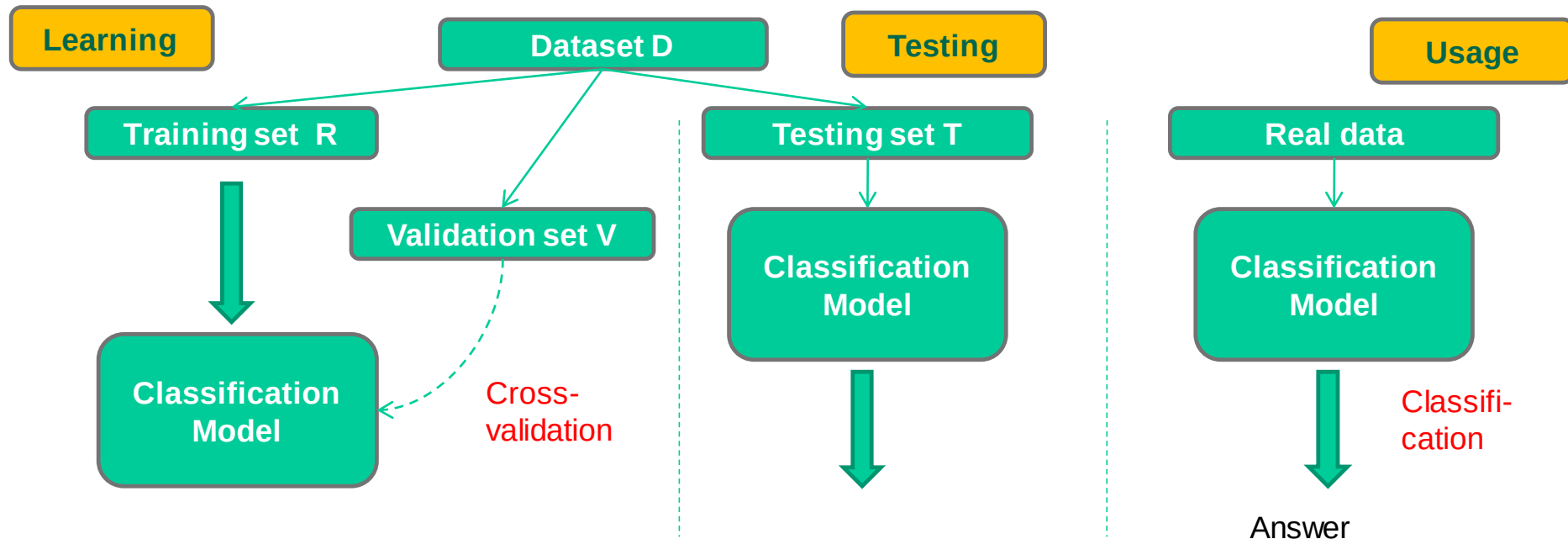
Basic concepts

Available information:

- set of labeled data:
 - $D = \{(x_1, c_1), (x_2, c_2), \dots, (x_N, c_N)\}$
 - Each x_i has n attributes (features)
 - The class label c_i belongs to $\{1, 2, \dots, K\}$

Goal:

- Based on data from D , design a classification model C such that
 - C can predict the class to which a new data x belongs



Classification models

A classification model is a “mapping” between attributes and class labels

Example of classification models:

- Decision trees
- Classification rules
- Prototypes (exemplars)
- Probabilistic models
- Function based models (e.g. support vector machines, neural networks etc.)

The classification model should be:

- Accurate:
 - Identify the right class
- Compact / comprehensible
 - Easy to be understood/ interpreted by the user (it is preferable to not be a black-box model)
- Efficient in the
 - Learning/training step
 - Classification step

Classification models

Example

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....
```

Simple classification rule:
IF (Cell_Size_Uniformity < 3.5)
THEN benign
ELSE malignant

Classification models

Example

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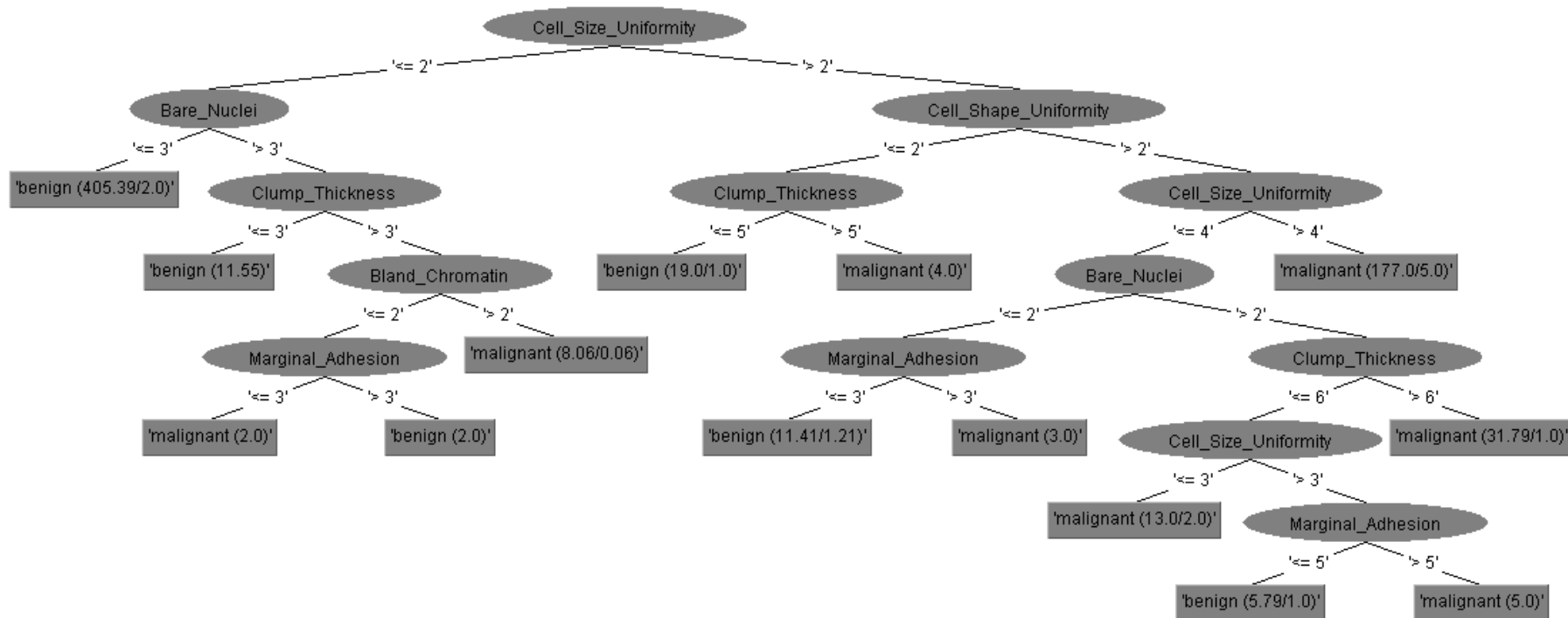
Simple classification rule:
IF (Cell_Size_Uniformity < 3.5)
THEN benign
ELSE malignant

Question: How good is this rule?

In 92.7% of cases the classifier produced the right class
How was this value computed?
How should it be interpreted?

Classification models

Example: a more complex model (decision tree)



Performance: in 94.56% of cases the classifier produced the right class

What about its readability?

Performance measures

Context: let us consider a 2-classes problem (binary classification)

- Class 1 – positive (e.g. malignant)
- Class 2 – negative (e.g. benign)

The simplest way to measure the performance is to analyze in how much cases the classifier provides the right answer – this information can be summarized in the so-called confusion matrix

Confusion matrix:

	C1	C2	← provided by the classifier
C1	TP	FN	
C2	FP	TN	

↑
true class

TP = True Positive = nr of cases which are in C1 and are correctly classified in C1

TN = True Negative = nr of cases which are in C2 and are correctly classified in C2

FP = False Positive = nr of cases which are in C2 but are wrongly classified in C1

FN = False Negative = nr of cases which are in C1 but are wrongly classified in C2

Performance measures

The case of K classes (multiple classification):

- One can construct a confusion matrix for each class (the current class is considered the positive one and all the other classes are grouped into an aggregated negative class)
- One can extend the confusion matrix to K classes by considering K rows and K columns

Confusion matrix:

	C_1	C_2	...	C_j	...	C_K
C_1	T_1	F_{12}	...	F_{1j}	...	F_{1K}
C_2	F_{21}	T_2	...	F_{2j}	...	F_{2K}
...	...	...	...	...	...	...
C_i	F_{i1}	F_{i2}	...	F_{ij}	...	F_{iK}
...	...	...	...	...	...	...
C_K	F_{K1}	F_{K2}	...	F_{Kj}	...	T_K

↑
true class

← provided by the classifier

T_1 = nr of cases that belong to C_1 and are correctly as being part of C_1

F_{ij} = nr of cases which should be classified in C_i but are classified in C_j

Performance measures

TP = True Positive = nr of cases which are in C1 and are correctly classified in C1

TN = True Negative = nr of cases which are in C2 and are correctly classified in C2

FP = False Positive = nr of cases which are in C2 but are wrongly classified in C1

FN = False Negative = nr of cases which are in C1 but are wrongly classified in C2

Accuracy = $(TP+TN)/(TP+TN+FP+FN)$ = ratio of correctly classified data

Sensitivity = $TP/(TP+FN)$ (TP rate or recall)

Specificity = $TN/(TN+FP)$ (TN rate), 1-specificity = $FP/(TN+FP)$ = FP rate

Precision = $TP/(TP+FP)$

Remarks:

- In all cases the value is in $[0,1]$; higher value suggest better performance
- Sensitivity and specificity are frequently used in medical data analysis while precision and recall are used in information retrieval

Performance measures

In the context of information retrieval:

Precision = $TP/(TP+FP)$ = $\text{card}(\text{relevant and retrieved}) / \text{card}(\text{retrieved})$

Recall = $TP/(TP+FN)$ = $\text{card}(\text{relevant and retrieved}) / \text{card}(\text{relevant})$

Both of them should be maximized; an aggregated variant is their harmonic mean

F-score = $2 * \text{precision} * \text{recall} / (\text{precision} + \text{recall})$

Remarks:

- F-score is frequently referred as F1-score, as it corresponds to $b=1$ for the weighted variant: $(1+b) * \text{precision} * \text{recall} / (b * \text{precision} + \text{recall})$
- a higher value of F-score corresponds to a better behavior

Performance measures

Robust measure for unbalanced datasets:

- Matthews Correlation Coefficient (MCC)

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

- MCC takes values in $[-1,1]$
 - $MCC = 1 \rightarrow$ perfect prediction
 - $MCC = 0 \rightarrow$ prediction similar to a random decision
 - $MCC = -1 \rightarrow$ total disagreement between prediction and correct classes

Performance measures

Cost sensitive accuracy

- In some cases (e.g. medical diagnosis) misclassifying the data belonging to one class may be more costly than misclassifying data belonging to the other class (e.g. missing a malignant case is more costly than missing a normal case) – in such cases the accuracy should take into account the cost
- $\text{CostAccuracy} = (\text{cost}_1 * n_1 * \text{sensitivity} + \text{cost}_2 * n_2 * \text{specificity}) / (\text{cost}_1 * n_1 + \text{cost}_2 * n_2)$
 - cost_i = cost of misclassifying data from C_i
 - n_i = number of data in C_i

Performance measures

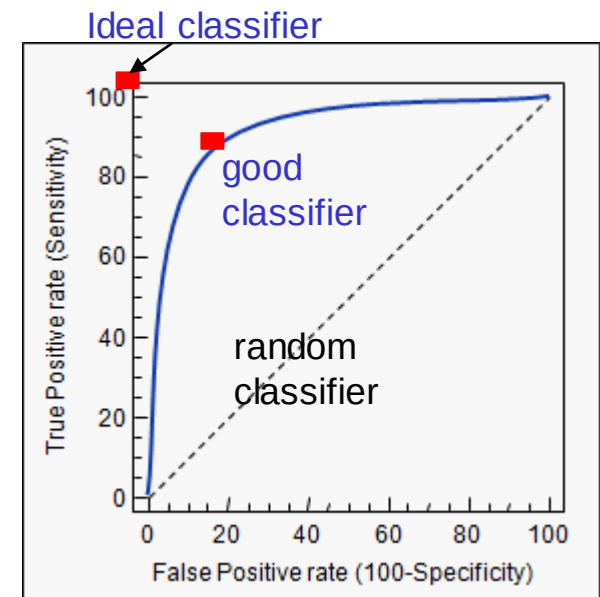
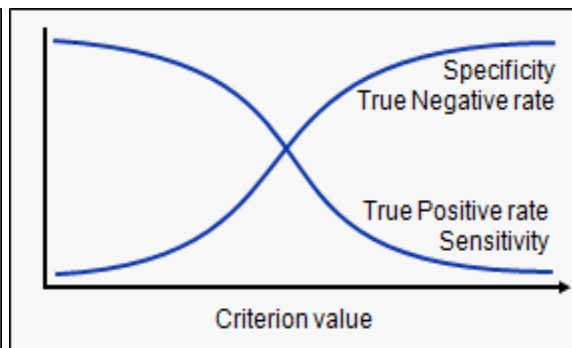
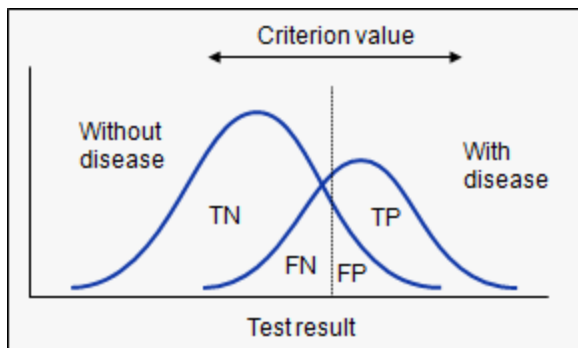
ROC Curve (Receiver Operator Characteristics) and Area Under Curve (AUC)

- Useful for evaluating binary classifiers which are based on a threshold
(ex: IF (Cell_Size_Uniformity < 3.5) THEN benign ELSE malignant)
- ROC curve = set of points having the coordinates
(FP rate, TP rate) = (1-specificity, sensitivity)

corresponding to different values of the threshold (or to different data subsets in the context of cross-validation) are joined in increasing order of FP rate:

(0,0), (FPR₁, TPR₁), (FPR₂, TPR₂), ... , (1,1)

(last point can be (100, 100) if percentages are used)



Source: <https://www.medcalc.org/manual/roc-curves.php>

The simplest classifier

Example:

- let us consider the dataset “sick” from UCI Machine Learning
- It contains records for 3772 patients, out of which:
 - 231 are sick (class C1 – positive) - 6%
 - 3541 are not sick (class C2 – negative) - 94%
- We are interested in a classifier having an accuracy at least equal to 0.9 (90%) for the available dataset
- Which is the simplest classifier which satisfies this requirement?

The simplest classifier

Example:

- let us consider the dataset “sick” from UCI Machine Learning
- It contains records with 29 attributes for 3772 patients, out of which:
 - 231 are sick (class C1 – positive) – 6%
 - 3541 are not sick (class C2 – negative) – 94%
- We are interested in a classifier having an accuracy at least equal to 0.9 (90%) for the available dataset
- Which is the simplest classifier which satisfy this requirement?
- By considering the rule: “disregarding the values of the attributes the class is C2 (negative)” we obtain that $\text{accuracy} = 3541/3772 = 0.94 > 0.9$
- Which is the value of the sensitivity? ($\text{sens} = TP/(TP+FN)$)
- Is this an appropriate classifier? Is such a classifier of any use?

The simplest classifier

- Is this an appropriate classifier? Is such a classifier of any use?
- This is the so-called **ZeroR** classifier which uses only the distribution of data over classes and always returns the label of the most frequent class in the data set (it relies on a simple voting mechanism)
- It is **not appropriate** as it will produce a wrong answer for all data in the less frequent classes
- However it can be used to establish a lower bound for the accuracy of a classifier: if a trained model performs **only slightly better than ZeroR**, it may not be effectively learning patterns

Back to performance evaluation

- Using the entire set of available data to construct the classifier is not a wise approach, as the classifier could **overfit** the training data:
 - It produces good results for data used in the learning process
 - It behaves poorly for data not used in the learning process
- A better approach is to split the available dataset in:
 - a **training subset** (used to construct the classifier)
 - a **testing subset** (used to estimate the performance)
- There are different splitting strategies and variants for sampling data to be included in the training and testing subsets

Remark:

Besides testing subset, another subset (**validation subset**) could be used in order to tune the parameters of the classifier

Back to performance evaluation

Splitting strategies:

- Holdout
 - Reserve 2/3 for training and 1/3 for testing
- Random subsampling
 - Repeated holdout (the performance is computed by averaging the results obtained for each repeat)
- Stratified sampling (useful in the case of unbalanced datasets)
 - oversampling vs undersampling
- Bootstrap
 - Sampling with replacement

Back to performance evaluation

- Cross validation
 - Split the dataset randomly into k disjoint subsets (the subsets are of the same size)
 - **k-fold**: train on $k-1$ partitions, test on the remaining one
 - **Leave-one-out**: $k=n$ (use $n-1$ data for training and one data for testing – useful when the dataset is small)

The overall performance is computed as the **average** of the performance measures computed in the k folds

Remarks:

- It is one of the most frequently used approach

One step beyond ZeroR

Dataset: [sick.arff](#), 29 attributes, 3772 instances (231 in class C1, 3541 in class C2), 2 classes

ZeroR (always classify into the C2): [accuracy=0.94](#)

OneR: construct classification rules which contains only one attribute in the left hand side

Inferred rules (via Weka OneR):

If $T3 < 0.25$ then C2 (negative)

If $T3$ in $[0.25, 0.35)$ then C1 (sick)

If $T3$ in $[0.35, 0.55)$ then C2 (negative)

If $T3$ in $[0.55, 1.15)$ then C1 (sick)

If $T3 \geq 1.15$ then C2 (negative)

If $T3$ value is missing then C2 (negative)

Accuracy: 0.96

OneR

Main idea: construct rules corresponding to each attribute and select the attribute **highest discriminative power** → the set of rules represents the classifier

Remark: it is appropriate for attributes which take discrete values

Algorithm:

FOR each attribute A_i do

FOR each value v_{ij} of A_i construct

R_{ij} : if $A_i = v_{ij}$ then class $C_{k(i,j)}$

(the majority class amongst the subset of instances having $A_i = v_{ij}$)

aggregate in a set of rules R_i corresponding to A_i and compute Err_i
(number of wrongly classified data)

ENDFOR

ENDFOR

Select the set of rules characterized by the smallest error

OneR

Example: weather/play dataset

Outlook: err=4

sunny: 2 yes/ 3 no ($\rightarrow$ no)

overcast: 4 yes/ 0 no ($\rightarrow$ yes)

rainy: 3 yes/2 no ($\rightarrow$ yes)

Relation: weather.symbolic

No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
3	overcast	hot	high	FALSE	yes
4	rainy	mild	high	FALSE	yes
5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
10	rainy	mild	normal	FALSE	yes
11	sunny	mild	normal	TRUE	yes
12	overcast	mild	high	TRUE	yes
13	overcast	hot	normal	FALSE	yes
14	rainy	mild	high	TRUE	no

OneR

Example: weather/play dataset

Outlook: err=4

sunny: 2 yes/ 3 no ($\rightarrow$ no)

overcast: 4 yes/ 0 no ($\rightarrow$ yes)

rainy: 3 yes/2 no ($\rightarrow$ yes)

Temperature: err=5

hot: 2 yes/2 no ($\rightarrow$ yes)

mild: 4 yes/2 no ($\rightarrow$ yes)

cool: 3 yes/ 1no ($\rightarrow$ yes)

Relation: weather.symbolic

No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
1	sunny	hot	high	FALSE	no
2	sunny	hot	high	TRUE	no
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5	rainy	cool	normal	FALSE	yes
6	rainy	cool	normal	TRUE	no
7	overcast	cool	normal	TRUE	yes
8	sunny	mild	high	FALSE	no
9	sunny	cool	normal	FALSE	yes
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OneR

Example: weather/play dataset

Outlook: err=4

sunny: 2 yes/ 3 no ($\rightarrow$ no)

overcast: 4 yes/ 0 no ($\rightarrow$ yes)

rainy: 3 yes/ 2 no ($\rightarrow$ yes)

Temperature: err=5

hot: 2 yes/ 2 no ($\rightarrow$ yes)

mild: 4 yes/ 2 no ($\rightarrow$ yes)

cool: 3 yes/ 1 no ($\rightarrow$ yes)

Humidity: err=5

high: 4 yes/ 4 no ($\rightarrow$ yes)

normal: 6 yes/ 1 no ($\rightarrow$ yes)

Relation: weather.symbolic

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OneR

Example: weather/play dataset

Outlook: err=4

sunny: 2 yes/ 3 no (→ no)

overcast: 4 yes/ 0 no (→ yes)

rainy: 3 yes/ 2 no (→ yes)

Temperature: err=5

hot: 2 yes/ 2 no (→ yes)

mild: 4 yes/ 2 no (→ yes)

cool: 3 yes/ 1 no (→ yes)

Humidity: err=5

high: 4 yes/ 4 no (→ yes)

normal: 6 yes/ 1 no (→ yes)

Windy: err=5

true: 3 yes/ 3 no (→ yes)

false: 6 yes/ 2 no (→ yes)

Relation: weather.symbolic

No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
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Rules: weather/play dataset

If outlook=sunny then “no”

If outlook=overcast then “yes”

If outlook=rainy then “yes”

Accuracy (training set): 0.71

Accuracy (cross-validation): 0.43 (!!)

OneR

Example: weather/play dataset

Rules: weather/play dataset

If outlook=sunny then “no”

If outlook=overcast then “yes”

If outlook=rainy then “yes”

Relation: weather.symbolic

No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
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Classification step:

- New day: (outlook=rainy, temperature=cool, humidity=high, windy=false)
- Answer: Yes

OneR

Summary of OneR implementation

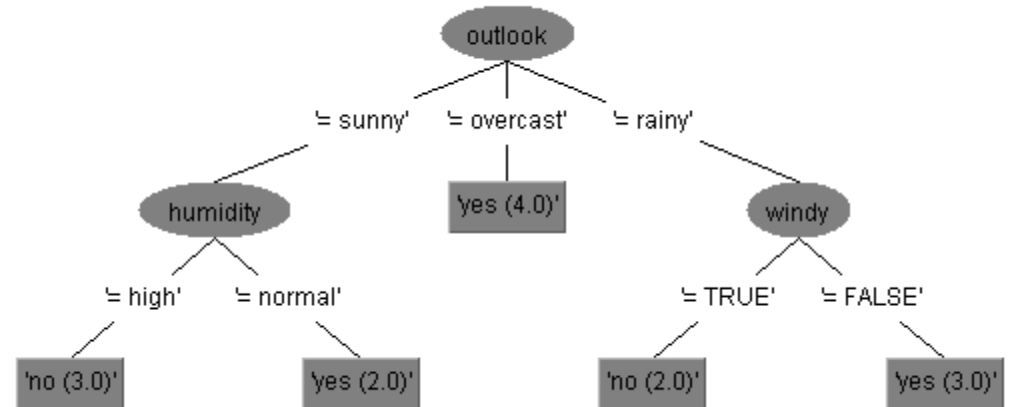
- **Construction of the set of rules (training step)**
 - **Input:** training set (labelled data instances)
 - **Output:** set of simple rules (all rules involve only one attribute – the same attribute in all rules)
 - **Algorithm:** search all attributes and corresponding values and select the attribute leading to the smallest classification error (see slide 26)
- **Using the set of rules (classification step)**
 - **Input:** set of rules, new data instance
 - **Output:** class label
 - **Algorithm:**
 - Find the rule which match the input data
 - Return the class corresponding to the matching rule

One step beyond OneR

Weather/play dataset

Relation: weather.symbolic					
No.	outlook Nominal	temperature Nominal	humidity Nominal	windy Nominal	play Nominal
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Decision tree (constructed using Weka)



How can be used?

Which class corresponds to a new instance:

(outlook=sunny, temperature=mild, humidity=normal, windy=False)?

Summary

- Construction of classification models
 - It requires labelled data (supervised training)
 - Dataset split: training subset + validation subset + testing subset
- Performance measures
 - Confusion matrix (true positives/negatives, false positive/negatives)
 - Accuracy, sensitivity (recall), specificity, precision
 - Aggregation of measures: F-score, AUC (area under the receiver operating curve)
- Interpretable models: decision trees, classification rules

Next lectures

- Decision trees
- Rule-based classifiers
- Instance based classifiers
 - kNN – nearest neighbor
- Probabilistic Classification
 - Naïve Bayes